Additional Status Codes

By default, **FastAPI** will return the responses using a `JSONResponse`, putting the content you return from your *path operation* inside of that `JSONResponse`.

It will use the default status code or the one you set in your *path operation*.

Additional status codes

If you want to return additional status codes apart from the main one, you can do that by returning a `Response` directly, like a `JSONResponse`, and set the additional status code directly.

For example, let's say that you want to have a *path operation* that allows to update items, and returns HTTP status codes of `2 0 0` "OK" when successful.

But you also want it to accept new items. And when the items didn't exist before, it creates them, and returns an HTTP status code of `2 0 1` "Created".

To achieve that, import `JSONResponse`, and return your content there directly, setting the `status_code` that you want:

Python 3.10+

```python
from typing import Annotated

from fastapi import Body, FastAPI, status
from fastapi.responses import JSONResponse

app = FastAPI()

items = {"foo": {"name": "Fighters", "size": 6}, "bar": {"name": "Tenders", "size": 3}}


@app.put("/items/{item_id}")
async def upsert_item(
    item_id: str,
    name: Annotated[str | None, Body()] = None,
    size: Annotated[int | None, Body()] = None,
):
    if item_id in items:
        item = items[item_id]
        item["name"] = name
        item["size"] = size
        return item
    else:
        item = {"name": name, "size": size}
        items[item_id] = item
        return JSONResponse(status_code=status.HTTP_201_CREATED, content=item)
```

Ask AI

Other versions and variants

Python 3.10+ - non-Annotated

Tip

Prefer to use the `Annotated` version if possible.

```python
from fastapi import Body, FastAPI, status
from fastapi.responses import JSONResponse

app = FastAPI()

items = {"foo": {"name": "Fighters", "size": 6}, "bar": {"name": "Tenders", "size": 3}}


@app.put("/items/{item_id}")
async def upsert_item(
    item_id: str,
    name: str | None = Body(default=None),
    size: int | None = Body(default=None),
):
    if item_id in items:
        item = items[item_id]
        item["name"] = name
        item["size"] = size
        return item
    else:
        item = {"name": name, "size": size}
        items[item_id] = item
        return JSONResponse(status_code=status.HTTP_201_CREATED, content=item)
```

Warning

When you return a `Response` directly, like in the example above, it will be returned directly.

It won't be serialized with a model, etc.

Make sure it has the data you want it to have, and that the values are valid JSON (if you are using `JSONResponse`).

Technical Details

You could also use `from starlette.responses import JSONResponse`.

FastAPI provides the same `starlette.responses` as `fastapi.responses` just as a convenience for you, the developer. But most of the available responses come directly from Starlette. The same with `status`.

OpenAPI and API docs

If you return additional status codes and responses directly, they won't be included in the OpenAPI schema (the API docs), because FastAPI doesn't have a way to know beforehand what you are going to return.

But you can document that in your code, using: Additional Responses.